

Subject: Submission of Manuscript: “AutoAntibiotic: A Reproducible Virtual Screening Pipeline Identifies Stable Rigid-Docking Poses for MRSA PBP2a That Require Extended MD for Confirmation”

Dear Editor,

We submit for consideration our manuscript entitled “AutoAntibiotic: A Reproducible Virtual Screening Pipeline for MRSA PBP2a with Preliminary Validation” for publication in the Journal of Chemical Information and Modeling.

This work presents the AutoAntibiotic Discovery Pipeline v7.4.0, a fully automated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers using AutoDock Vina with mechanism-restricted selectivity profiling against human serine hydrolases, MMFF94-based rescoring, and OpenMM complex minimisation and short MD for top candidates. The pipeline additionally supports flexible side-chain docking and an MD stability filter.

Key findings include:

- A validated protocol with core RMSD of 1.251 Å (Validated badge)
- Enrichment validation FAILS: a standardised DUD-E-style benchmark (76 ChEMBL actives, 369 property-matched decoys, apo 1VQQ, exhaustiveness 8) gives $AUC = 0.148$ and $EF_{1\%} = 0.00$ (below the PASS threshold of $AUC \geq 0.7$ and $EF_{1\%} \geq 5.0$); the water-included receptor benchmark gives $AUC = 0.143$. Both reflect the limited capacity of rigid-receptor docking to rank weak PBP2a actives above property-matched decoys for this challenging target
- Two lead compounds with $SI \geq 2.0$ (Strong tier): ALL_QU04 (primary lead, $SI = 2.07$, MMFF94 strain = 366 a.u., $SA = 1.78$) and BRICS_0022 (secondary, $SI = 2.13$, strain = 738 a.u.), both engaging all three catalytic residues (Ser403, Lys406, Tyr446)
- Two Promising-tier sulfonamide scaffolds ALL_SU08 ($SI = 1.91$, $SA = 1.81$) and ALL_SP03 ($SI = 1.93$, $SA = 3.27$) with confirmed catalytic-triad contacts
- OpenMM gas-phase minimisation confirmed the docked poses are local minima, and short MD – with the ligand correctly placed in the docked pose – showed the top candidates remaining bound over the simulated windows (explicit-solvent 100 ps NPT ligand RMSD 1.86–3.49 Å, with Ser403 H-bond occupancy 1.00 for BRICS_0022 and SEED_01150). These short-timescale results provide preliminary support for the docked binding modes; extended multi-replica unrestrained MD and induced-fit docking remain required for long-term stability confirmation

The manuscript describes a rigorous, reproducible methodology with transparent validation criteria and a candid discussion of limitations, including the sensitivity of enrichment

benchmarks to methodological choices. We believe this work will be of interest to the journal's readership working on antibiotic discovery and structure-based drug design.

All authors have approved the manuscript and agree with its submission to your journal.

Sincerely,

The AutoAntibiotic Agent
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